

TARGET BUSINESS CASE

1. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

Q1(A). Data type of all columns in the "customers" table.

Answer:

```
SELECT column_name, data_type
```

```
FROM `scaler-dbms-sql-452714.target.INFORMATION_SCHEMA.COLUMNS`
```

```
WHERE table_name = 'customers';
```

Query Result:

Job information			Results	Chart	JSON	Execution
Row	column_name	data_type				
1	customer_id	STRING				
2	customer_unique_id	STRING				
3	customer_zip_code_prefix	INT64				
4	customer_city	STRING				
5	customer_state	STRING				

Insights:

- The following table contains information related to customers, mostly in STRING form.

Recommendations: N/A

Q1(B). Get the time range between which the orders were placed.

Answer:

```
SELECT min(order_purchase_timestamp) as first_order, max(order_purchase_timestamp) as  
last_order, timestamp_diff(max(order_purchase_timestamp),  
min(order_purchase_timestamp), hour) as time_difference
```

```
FROM `target.orders`;
```

Query Result:

Job information				Results	Chart	JSON	Execution details	Exe
Row	first_order	last_order	time_difference					
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	18548					

Insights:

- The orders were placed between 2016-09-04 (21:15:19 UTC) and 2018-10-17 (17:30:18 UTC), covering a total of 18,548 hours or approximately 772 days.

Recommendations: N/A

Q1(C). Count the Cities & States of customers who ordered during the given period.

Answer:

```
SELECT count (distinct c.customer_city) as cities, count (distinct c.customer_state) as states
FROM `target.customers` c
JOIN `target.orders` o
ON c.customer_id = o.customer_id;
```

Query Result:

Job information				Results	Chart
Row	cities	states			
1	4119	27			

Insights:

- The customers who placed the orders came from 4,119 different cities across 27 different states indicating wide geographic stretch and a diverse customer base.

Recommendations:

- There is still room for expansion into the untapped market of the cities where the company is not yet operational.

2. In-depth Exploration:

Q2(A). Is there a growing trend in the no. of orders placed over the past years?

Answer:

```
SELECT EXTRACT (year from order_purchase_timestamp) as years, COUNT (order_id)
number_of_orders_placed
```

```
FROM `target.orders`
```

```
GROUP BY 1
```

```
ORDER BY 1;
```

Query Result:

Job information		Results	Chart	JSON
Row	years	number_of_orders_placed		
1	2016	329		
2	2017	45101		
3	2018	54011		

Insights:

- There is a consistent growth in the number of orders being placed, showing an upward trend in customer engagement and usage of the platform.

Recommendations:

- Sustain and accelerate this growth by retaining the present customers but also increasing them.
- Invest in infrastructure, plan for the inventory, and logistics for the continued growth.
- Analyse why there was a steep increase in 2017, and try to replicate those to gain more customers.

Q2(B). Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Answer:

```
SELECT EXTRACT (year from order_purchase_timestamp) as years, EXTRACT (month from
order_purchase_timestamp) as months, COUNT (order_id) number_of_orders_placed
```

```
FROM `target.orders`
```

```
GROUP BY 1,2
```

```
ORDER BY 1,2;
```

Query Result:

Job information		Results	Chart	JSON	Execution details
Row	years	months	number_of_orders_placed		
1	2016	9	4		
2	2016	10	324		
3	2016	12	1		
4	2017	1	800		
5	2017	2	1780		
6	2017	3	2682		
7	2017	4	2404		
8	2017	5	3700		
9	2017	6	3245		
10	2017	7	4026		

Insights:

- There is a consistent growth in number of orders placed, particularly when the numbers peaked, i.e. in late 2017 and early 2018.
- But there is also a drastic decline in September and October of 2018.

Recommendations:

- Analyse what worked (promotions, advertisements, offers, campaigns, etc.) that led to the rise in numbers and plan the logistics, inventory, servers accordingly for the future growth and actions.
- Also investigate the reason for a drastic decline in orders in late 2018, whether there is missing data entry, tracking issues, or actual business decline.

Q2(C). During what time of the day, do the Brazilian customers mostly place their orders?
(Dawn, Morning, Afternoon or Night)

Answer:

```
SELECT case when extract (hour from order_purchase_timestamp) between 0 and 6 then  
'Dawn'
```

```
when extract (hour from order_purchase_timestamp) between 7 and 12 then 'Morning'
```

```
when extract (hour from order_purchase_timestamp) between 13 and 18 then 'Afternoon'
```

```
when extract (hour from order_purchase_timestamp) between 19 and 23 then 'Night' end  
as time_of_day,
```

```
COUNT (order_id) as total_orders
```

```
FROM `target.orders`

GROUP BY time_of_day

ORDER BY total_orders desc;
```

Query Result:

Job information		Results	Chart	JSON
Row	time_of_day	total_orders		
1	Afternoon	38135		
2	Night	28331		
3	Morning	27733		
4	Dawn	5242		

Insights:

- Most of the orders are placed in afternoon, followed by that in night and then in the morning. Very few orders were placed in the dawn.

Recommendations:

- Target afternoon slot with time sensitive deals and offers.
- Try to make night and morning slots more engaging by sending notifications, or time specific promos.
- As dawn saw the least number of orders, there is little incentive to spend on this time slot.

3. Evolution of E-commerce orders in the Brazil region:

Q3(A). Get the month on month no. of orders placed in each state

Answer:

```
SELECT c.customer_state, extract (month from o.order_purchase_timestamp) as months,
count(o.order_id) as no_of_orders
```

```
FROM `target.orders` o
```

```
left join `target.customers` c
```

```
on o.customer_id = c.customer_id
```

```
group by 1,2
```

```
order by 1,2;
```

Query Result:

Job information <u>Results</u> Chart JSON Execution detail				
Row	customer_state ▾	months ▾	no_of_orders ▾	
1	AC	1	8	
2	AC	2	6	
3	AC	3	4	
4	AC	4	9	
5	AC	5	10	
6	AC	6	7	
7	AC	7	9	
8	AC	8	7	
9	AC	9	5	
10	AC	10	6	

Insights:

- The highest number of orders in any month, are usually from the state of SP (Sao Paulo), followed by RJ (Rio de Janeiro) and MG (Minas Gerais).
- The lowest number of monthly orders were usually from the states of RR (Roraima), and AP (Amapa).

Recommendations:

- Focus on high performing states, and try to retain the already ordering customers by offering them 'monthly subscription' based plans.
- Analyse the barriers of the low performing states, like lack of awareness, internet connection, delivery issues, etc.

Q3(B). How are the customers distributed across all the states?

Answer:

```
SELECT distinct customer_state, COUNT (distinct customer_id) as no_of_customers
from `target.customers`
group by 1
order by 2 desc;
```

Query Result:

Job information		Results	Chart	JSON
Row	customer_state	no_of_customers		
1	SP	41746		
2	RJ	12852		
3	MG	11635		
4	RS	5466		
5	PR	5045		
6	SC	3637		
7	BA	3380		
8	DF	2140		
9	ES	2033		
10	GO	2020		

Insights:

- The highest number of customers are from the state SP (Sao Paulo) and that too from a large margin. The next two biggest customer base are in the state of RJ (Rio de Janeiro) and MG (Minas Gerais).
- The customer state RR (Roraima) and AP (Amapa) has the lowest number of customers in the country.

Recommendations:

- Use the high performing states as a testing ground for new features, products and services, and try to optimize the logistics for further expansion into other states.
- Analyse the barriers of the low performing states, like lack of awareness, internet connection, delivery issues, etc.

4. Impact on Economy: Analyse the money movement by e-commerce by looking at order prices, freight and others.

Q4(A). Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

Answer:

```

WITH yearly AS
(SELECT order_id,
      EXTRACT (year from order_purchase_timestamp) as years,
      EXTRACT (month from order_purchase_timestamp) as months
FROM `target.orders`
where EXTRACT (year from order_purchase_timestamp) in (2017,2018)
and EXTRACT (month from order_purchase_timestamp) between 1 and 8),

```

```

final as
(SELECT y.years,
      sum(p.payment_value) as total
FROM yearly y
join `target.payments` p
on y.order_id = p.order_id
group by years)

```

```

SELECT max (case when years = 2017 then total end) as year_2017,
      max (case when years = 2018 then total end) as year_2018,
      round (((max (case when years = 2018 then total end) - max (case when years = 2017
then total end)) / max (case when years = 2017 then total end)) * 100,2) as inc_percent
from final;

```

Query Result:

Job information Results Chart JSON Exec				
Row	year_2017	year_2018	inc_percent	
1	3669022.119999...	8694733.840000...	136.98	

Insights:

- The year 2018 showed an increase of 136.98% in cost of orders, compared to 2017
- The data takes into account only performance from January to August of both years.

Recommendation:

- Sustain the growth rate to perform better in the next year of 2019.

Q4(B). Calculate the Total & Average value of order price for each state.

Answer:

```
SELECT c.customer_state,  
       sum(payment_value) as total_value,  
       avg(payment_value) as avg_value  
FROM `target.payments` p  
JOIN `target.orders` o  
ON o.order_id = p.order_id  
JOIN `target.customers` c  
ON c.customer_id = o.customer_id  
GROUP BY c.customer_state  
ORDER BY 2 desc,3 desc;
```

Query Result:

Job information		Results	Chart	JSON	Execution detail
Row	customer_state ▼	total_value ▼	avg_value ▼		
1	SP	5998226.960000...	137.5046297739...		
2	RJ	2144379.689999...	158.5258882235...		
3	MG	1872257.259999...	154.7064336473...		
4	RS	890898.5400000...	157.1804057868...		
5	PR	811156.3800000...	154.1536259977...		
6	SC	623086.4300000...	165.9793367075...		
7	BA	616645.8200000...	170.8160166204...		
8	DF	355141.0799999...	161.1347912885...		
9	GO	350092.3100000...	165.7634043560...		
10	ES	325967.5499999...	154.7069530137...		

Insights:

- Highest value of order price was from the state SP (Sao Paulo).

- The only states apart from SP that had total value in 7 digits were RJ and MG.
- The highest average value of order price was from the state PB (Paraiba)
- The state of SP (Sao Paulo) ranked the lowest in average value of order price, which means people in SP maybe buying or ordering more as they rank highest in total value, but are buying items that are not that expensive.

Recommendation:

- As SP (Sao Paulo) have high number of orders but low average cost per order, introduce strategies like bundling and upselling to increase average order value.
- Invest more in states like of RJ (Rio de Janeiro) and MG (Minas Gerais) which have a high potential for future growth as they are already performing well.
- As PB (Paraiba) has highest average value, try to increase the customer frequency or volume there but still maintain the average value.

Q4(C). Calculate the Total & Average value of order freight for each state.

Answer:

```
SELECT c.customer_state,
       round(sum(i.freight_value),2) as total,
       round(avg(i.freight_value),2) as average
FROM `target.order_items` i
JOIN `target.orders` o
ON o.order_id = i.order_id
JOIN `target.customers` c
ON c.customer_id = o.customer_id
GROUP BY c.customer_state
ORDER BY 2 desc,3 desc;
```

Query Result:

Job information		Results	Chart	JSON	Execution detail
Row	customer_state ▼	total ▼	average ▼		
1	SP	718723.07	15.15		
2	RJ	305589.31	20.96		
3	MG	270853.46	20.63		
4	RS	135522.74	21.74		
5	PR	117851.68	20.53		
6	BA	100156.68	26.36		
7	SC	89660.26	21.47		
8	PE	59449.66	32.92		
9	GO	53114.98	22.77		
10	DF	50625.5	21.04		

Insights:

- States of SP (Sao Paulo), RJ (Rio de Janeiro) and MG (Minas Gerais) amounts to a significant volume of total freight. This indicates these states have higher economic activity, large population, and good transportation leading to less freight value.
- States of RR (Roraima), PB (Paraiba) RO (Rondonia) and AC (Acre) have highest average of freight value suggesting that shipment cost is high because of distance from metropolitan cities, poor infrastructure or less amount of shipments.

Recommendation:

- Improve efficiency for high volume in states of SP, RJ and MG by methods like route optimization, warehouse management, and logistics.
- In state having high average freight value, focus on cost efficiency and how to reduce the cost for shipment.

5. Analysis based on sales, freight and delivery time.

Q5(A). Find the no. of days taken to deliver each order from the order's purchase date as delivery time.

Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

Answer:

```
SELECT order_id,  
date(order_purchase_timestamp) as purchase_date,  
date(order_delivered_customer_date) as delivery_date,  
date(order_estimated_delivery_date) as delivery_estimate,  
timestamp_diff(order_delivered_customer_date, order_purchase_timestamp, day) as  
time_to_deliver,  
timestamp_diff(order_delivered_customer_date, order_estimated_delivery_date, day) as  
diff_estimated_delivery  
FROM `target.orders`  
  
where order_purchase_timestamp is not null  
and order_delivered_customer_date is not null  
and order_estimated_delivery_date is not null  
order by 5 desc;
```

Query Result:

Job information		Results	Chart	JSON	Execution details		Execution graph	
Row	order_id	purchase_date	delivery_date	delivery_estimate	time_to_deliver	diff_estimated_delivery		
1	ca07593549f1816d26a572e06...	2017-02-21	2017-09-19	2017-03-22	209	181		
2	1b3190b2dfa9d789e1f14c05b6...	2018-02-23	2018-09-19	2018-03-15	208	188		
3	440d0d17af552815d15a9e41a...	2017-03-07	2017-09-19	2017-04-07	195	165		
4	285ab9426d6982034523a855f...	2017-03-08	2017-09-19	2017-04-06	194	166		
5	0f4519c5f1c541ddec9f21b3bd...	2017-03-09	2017-09-19	2017-04-11	194	161		
6	2fb597c2f772eca01b1f5c561bf...	2017-03-08	2017-09-19	2017-04-17	194	155		
7	47b40429ed8cce3aee9199792...	2018-01-03	2018-07-13	2018-01-19	191	175		
8	2fe324feb907e3ea3f2aa96508...	2017-03-13	2017-09-19	2017-04-05	189	167		
9	2d7561026d542c8dbd8f0daea...	2017-03-15	2017-09-19	2017-04-13	188	159		
10	c27815f7e3dd0b926b5855262...	2017-03-15	2017-09-19	2017-04-10	187	162		

Insights:

- The highest number of days an order took to deliver, compared to its ordering date was 209 and 208 days. The delayed orders were as late as 5 months.
- More than 1,500 orders were delivered in 1 day which were mostly before the estimated delivery date.
- The difference between estimated delivery and actual delivery was as high as 5 to 6 months. But some deliveries were also made as early from the estimated delivery as 4 to 5 months.

Recommendations:

- Check if the reason for the delay in order delivery is warehouse, shipping, or labelling issues.
- If there is high estimated delivery, update the customer time to time to reduce customer frustration and anxiety.
- If the reason for delay is the location of the place, invest in local small warehouses so that sourcing could be easy.

Q5(B). Find out the top 5 states with the highest & lowest average freight value.

Answer:

WITH avg_freight_value as

(SELECT c.customer_state,

round(avg(i.freight_value),2) as average

FROM `target.order\_items` i

JOIN `target.orders` o

ON o.order_id = i.order_id

JOIN `target.customers` c

ON c.customer_id = o.customer_id

GROUP BY c.customer_state),

highest_freight as

(SELECT customer_state,average,'highest 5' as rank_type

FROM avg_freight_value

ORDER BY average desc

LIMIT 5),

lowest_freight as

(SELECT customer_state,average,'lowest 5' as rank_type

FROM avg_freight_value

ORDER BY average

LIMIT 5)

SELECT *

FROM highest_freight

UNION ALL

SELECT *

FROM lowest_freight;

Query Result:

Job information		Results	Chart	JSON	Execution details	Exe
Row	customer_state	average	rank_type			
1	RR	42.98	highest 5			
2	PB	42.72	highest 5			
3	RO	41.07	highest 5			
4	AC	40.07	highest 5			
5	PI	39.15	highest 5			
6	SP	15.15	lowest 5			
7	PR	20.53	lowest 5			
8	MG	20.63	lowest 5			
9	RJ	20.96	lowest 5			
10	DF	21.04	lowest 5			

Insights:

- States of RR, PB, RO, AC and PI have high freight value indicating the geographical location, that might be making it difficult for transportation and logistics.
- On the other side, SP, PR, MG, RJ and DF have low freight value, which mean they are economically developed states, thus having better logistics, order delivery optimization, better warehouse management and better transport connectivity.
- The gap between states with high freight value and low freight value also indicates the economic regional disparity in logistics and cost efficiency.

Recommendation:

- Invest in small warehouses for far flung areas so that the freight value could be cutdown because of less transportation cost.

- Invest in local freight carriers in the region so that they can be well trained and better equipped to handle the logistics.
- By having the low freight states as the distribution hubs, promote multi modal transportation that can cut the cost down for the delivery to regions that have high freight.

Q5(C). Find out the top 5 states with the highest & lowest average delivery time.

Answer:

WITH delivery_time as

(SELECT order_id,

customer_id,

date(order_purchase_timestamp) as purchase_date,

date(order_delivered_customer_date) as delivery_date,

timestamp_diff(order_delivered_customer_date, order_purchase_timestamp, day) as
time_to_deliver

FROM `target.orders`

where order_purchase_timestamp is not null

and order_delivered_customer_date is not null),

highest_delivery as

(SELECT customer_state,

round(avg(time_to_deliver),4) as avg_delivery,

'highest 5' as rank_type

from delivery_time d

join `target.customers` c

on c.customer_id = d.customer_id

group by customer_state

order by 2 desc

limit 5),

```

lowest_delivery as
(SELECT customer_state,
       round(avg(time_to_deliver),4) as avg_delivery,
       'lowest 5' as rank_type
from delivery_time d
join `target.customers` c
on c.customer_id = d.customer_id
group by customer_state
order by 2
limit 5)

```

```

select *
from highest_delivery
union all
select *
from lowest_delivery;

```

Query Result:

Job information		Results	Chart	JSON	Execution details	Exe
Row	customer_state ▼	avg_delivery ▼	rank_type ▼			
1	RR	28.9756	highest 5			
2	AP	26.7313	highest 5			
3	AM	25.9862	highest 5			
4	AL	24.0403	highest 5			
5	PA	23.3161	highest 5			
6	SP	8.2981	lowest 5			
7	PR	11.5267	lowest 5			
8	MG	11.5438	lowest 5			
9	DF	12.5091	lowest 5			
10	SC	14.4796	lowest 5			

Insights:

- The states having high average delivery time are mostly states that are far away from country's economic hubs.
- Poor infrastructure, geographical barriers, and a great distance from major cities causes a delay from the time of dispatch to the order actually reaching those areas.
- For the states with low average delivery time, they are mostly states with major cities, thus having large population enabling frequent shipments and faster deliveries.
- As these regions (with low delivery time) are major population centres, they are also an economic hub thus benefiting from the infrastructure and better logistics that comes with it.

Recommendation:

- Cooperate with the local governments for better infrastructure by investing in local regional warehouses, making the government having an incentive to push for better infrastructure in these remote areas.
- Try using unconventional modes of transportation for quick delivery, like river cargos.

Q5(D). Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

Answer:

```
SELECT c.customer_state,  
  
ROUND (avg(timestamp_diff(order_delivered_customer_date,  
order_estimated_delivery_date, day)),4) as diff_estimated_delivery  
  
FROM `target.orders` o  
  
JOIN `target.customers` c  
  
on o.customer_id = c.customer_id  
  
where order_delivered_customer_date is not null  
  
and order_estimated_delivery_date is not null  
  
group by c.customer_state  
  
order by diff_estimated_delivery  
  
limit 5
```

Query Result:

Job information		Results	Chart	JSON
Row	customer_state		diff_estimated_d...	
1	AC		-19.7625	
2	RO		-19.1317	
3	AP		-18.7313	
4	AM		-18.6069	
5	RR		-16.4146	

Insights:

- The negative result meant the delivery was faster, which means before the estimated date.
- States of AC (Acre), RO(Rondonia), AP(Amapa), AM(Amazonas) and RR (Roraima) are states having fast delivery on average as compared to their estimate delivery time.
- This maybe because of the reason as these regions are not actually that connected through good transportation so the delivery date is overestimated.

Recommendations:

- Although early delivery is good, but there needs to be a balance of expected date and surprise as customer may keep expectations that may not be fulfilled always.
- We need to check if the early deliveries are because of some premium couriers or transportation routes or modes, that maybe be costing the company.
- But overall the early delivery is a good sign of logistics and delivery reliability of the company that needs to be appreciated.

6. Analysis based on the payments:

Q6(A). Find the month on month no. of orders placed using different payment types.

Answer:

```
SELECT extract (month from order_purchase_timestamp) as months,  
       payment_type,  
       count(o.order_id) as no_of_orders
```

```

FROM `target.orders` o
JOIN `target.payments` p
ON p.order_id = o.order_id
GROUP BY 1,2
ORDER BY 1,3 desc;

```

Query Result:

Job information		Results	Chart	JSON	Execution details
Row	months ▾	payment_type ▾	no_of_orders ▾		
1	1	credit_card	6103		
2	1	UPI	1715		
3	1	voucher	477		
4	1	debit_card	118		
5	2	credit_card	6609		
6	2	UPI	1723		
7	2	voucher	424		
8	2	debit_card	82		
9	3	credit_card	7707		
10	3	UPI	1942		

Insights:

- The most prominent method of payment constituting majority of payments in credit card, usually ranking highest in the most used method of payment.
- The second most used method of payment is UPI, which shows that the country is adopting to this new method of digitisation, but it is still not enough as compared to credit card.
- Voucher as a payment method ranked higher than debit card in monthly usings.

Recommendations:

- As most people are buying using credit card, try to incentives purchases by credit card and UPI by offering rewards.
- Analyse the decline of vouchers and debit card as method of payment.

Q6(B). Find the no. of orders placed on the basis of the payment installments that have been paid.

Answer:

```
SELECT payment_installments,  
       count (distinct order_id) as no_of_order  
FROM `target.payments`  
GROUP BY payment_installments  
ORDER BY payment_installments;
```

Query Result:

Job information		Results	Chart
Row	payment_installm...	no_of_order	
1	0	2	
2	1	49060	
3	2	12389	
4	3	10443	
5	4	7088	
6	5	5234	
7	6	3916	
8	7	1623	
9	8	4253	
10	9	644	

Insights:

- There is as high drop after first instalment of nearly 75% to the second instalment suggesting many customers pay first instalment and drop off.
- From 2nd instalment up until 9th instalment, there is a gradual decline, but in 10th instalment there is a sudden rise.
- Very few orders are beyond 12th instalment up until 24th showing there is very less engagement in that segment.

Recommendation:

- Design instalment plans based on behaviour.
- Offer incentives like cashback for customers crossing third instalments.
- Since not many people crossed 12th instalment, design the payment structure accordingly.